ONLINE MASTERS IN **DATA SCIENCE**

DSC258: NATURAL LANGUAGE PROCESSING

VECTOR SEMANTICS & EMBEDDINGS

UC San Diego

COMPUTER SCIENCE & ENGINEERING

HALICIOĞLU DATA SCIENCE INSTITUTE

RECAP

A typical text mining pipeline…

Preprocessing
- Try to make texts “better formatted”

Representation
- Make the text data machine-actionable

Model Training
- Specifically for the end-task

OUTLINE

Word Embedding

- **Traditional Distributional Representation**
- Dense Vector Representations
- Interesting Properties of Word Vectors
- Applying Word Embeddings

Vector Semantics

Input

- A word *w*

Output

- A **vector** representation $\vec{w}_d$
  - In a **multi-dimensional** semantic space
  - Derived from the distribution of **neighboring** words

We wish to preserve the word meaning as much as possible

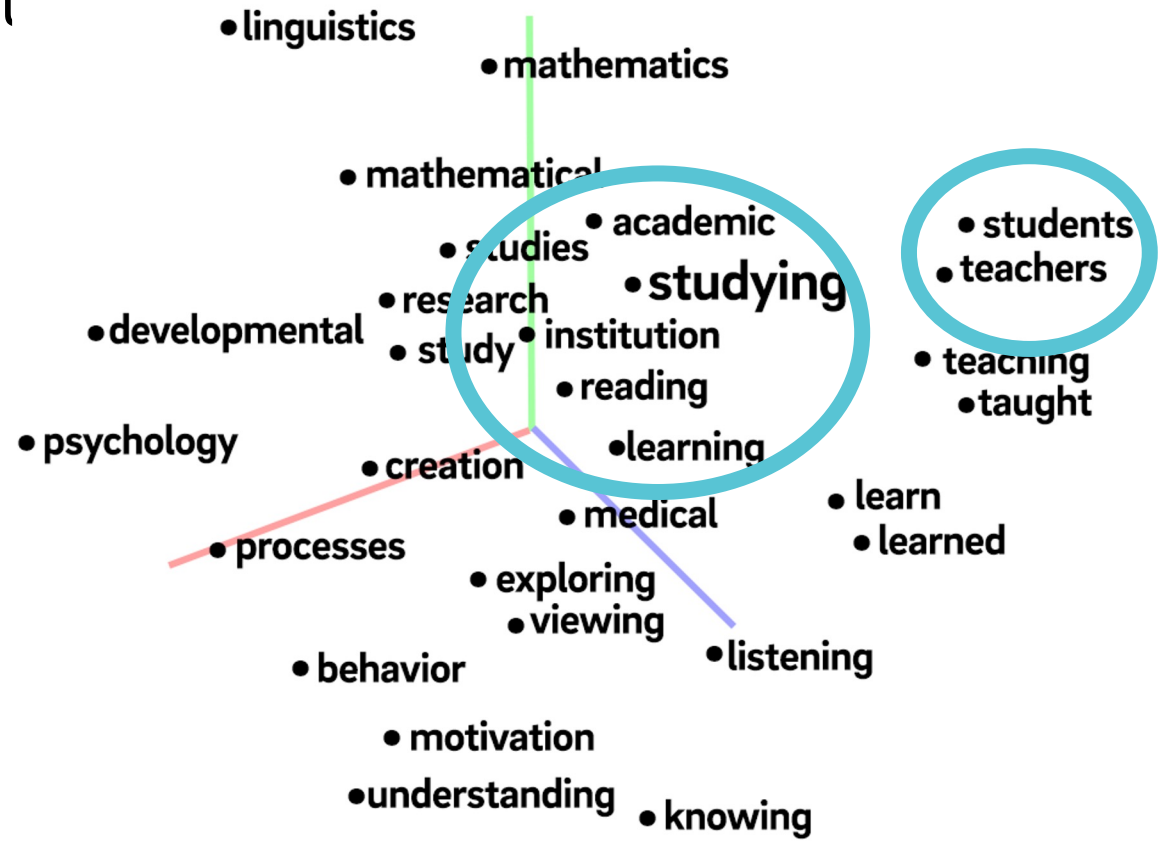
What is **tezgüino**?

- There is some ***tezgüino***.
- A jar of ***tezgüino*** is on the table.
- You need a lot of ***tezgüino*** to get your land cleared.
- Everybody likes ***tezgüino***.
- I'll have a drink of ***tezgüino***.
- ***Tezgüino*** makes you drunk.
- We make ***tezgüino*** out of corn, but we do not distill it.

From these seven contexts (distribution of neighboring words), one can determine that "tezgüino" is a kind of beer. (Lin, 1998, Nida, 1975)

Word Embedding Idea

Goal: Given a corpus, look for a latent space, which preserves the semantics as mu



Kastrati, Zenun, Arianit Kurti, and Ali Shariq Imran. "WET: Word embedding-topic distribution vectors for MOOC video lectures dataset." *Data in brief* 28 (2020): 105090.

Distributional Hypothesis

- John Firth, 1957: “You shall know a word by the company it keeps”
- Words which often have the same context words tend to be **similar** in semantics
- Examples of similar words (not synonym):
  - “dog” vs. “cat”
  - “coffee” vs. “tea”
  - …

Distributional Similarity

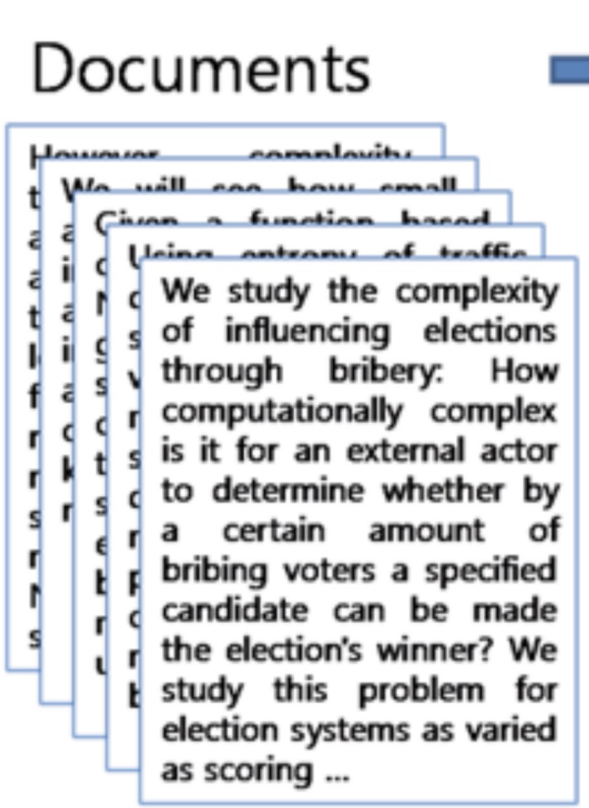
- A word *w* will be represented as a vector $\vec{\boldsymbol{w}} \in R^d$
- Distributional similarities use the set of contexts in which words appear to measure their similarity
- The similarity between words *u* and *v* can be decided by that between $\vec{\boldsymbol{u}}$ and $\vec{\boldsymbol{v}}$
  - Cosine similarity is a common practice

How to define “context”?

Option 1: Context = Document where the word appears

Matrix dimension: |V| X d

- V approx. 10-50K frequent
- d could be much larger



Vector-space representation

d = # of documents

	D1	D2	D3	D4	D5
complexity	2	0	3	2	3
algorithm	3	0	0	4	4
entropy	1	0	0	2	0
traffic	0	2	3	0	0
network	0	1	4	0	0

Term-document matrix

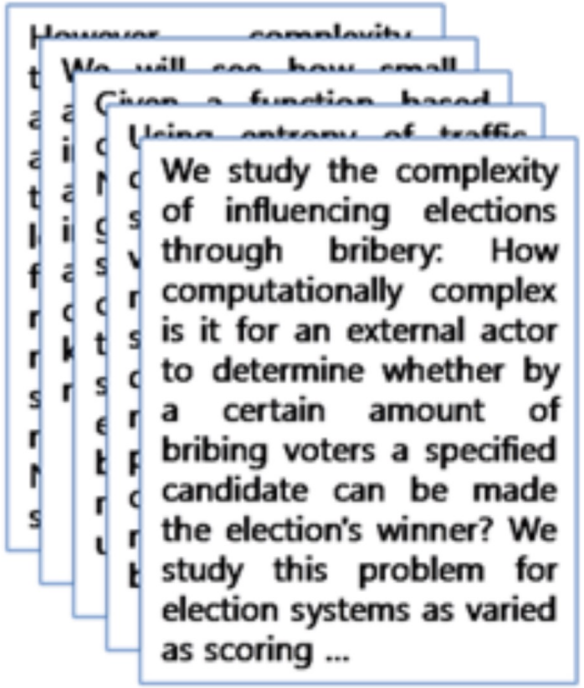
Image Source: https://bit.ly/3DcMFgn/

How to define “context”?

Option 1: Context = Document where the word appears

Raw count is skewed towards non-informative words e.g., “the”, “it”

Documents



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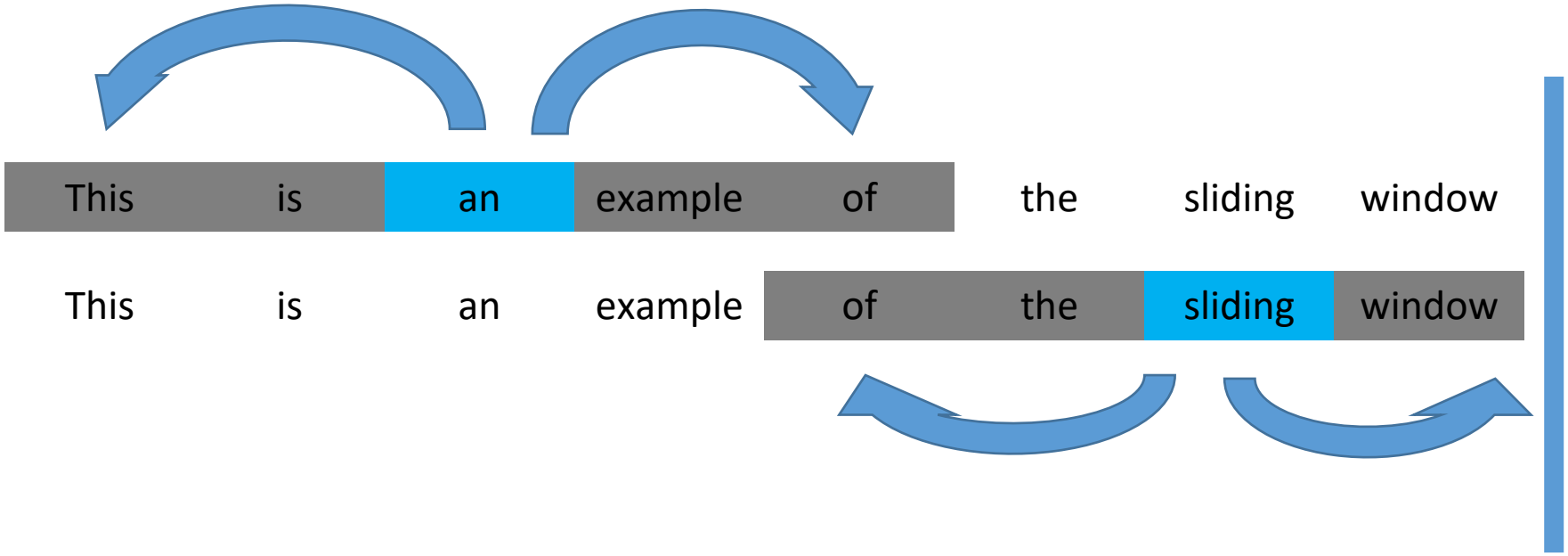
Term-Document Matrix

TF-IDF vectors could be a special case of this

How to define context?

Option 2: Context = nearby/co-occurrence

- **Within the same sliding window**
  - **E.g., window size = 5 (i.e., ± 2)**



How to define context?

Option 2: Context = nearby/co-occurrence

Term-Term Matrix

W =

I like deep learning. I like NLP. I enjoy flying.

Context Words

|V|

Target Words

|V|

counts	I	like	enjoy	deep	learning	NLP	flying	.
I	0	2	1	0	0	0	0	0
like	2	0	0	1	0	1	0	0
enjoy	1	0	0	0	0	0	1	0
deep	0	1	0	0	1	0	0	0
learning	0	0	0	1	0	0	0	1
NLP	0	1	0	0	0	0	0	1
flying	0	0	1	0	0	0	0	1
.	0	0	0	0	1	1	1	0

Source: https://bit.ly/3TGTaPH

How to define context?

Option 2: Context = nearby/co-occurrence

Term-Term Matrix

Resultant vectors tend to be sparse

Context Words |V|

Target Words |V|

counts	I	like	enjoy	deep	learning	NLP	flying	.
I	0	2	1	0	0	0	0	0
like	2	0	0	1	0	1	0	0
enjoy	1	0	0	0	0	0	1	0
deep	0	1	0	0	1	0	0	0
learning	0	0	0	1	0	0	0	1
NLP	0	1	0	0	0	0	0	1
flying	0	0	1	0	0	0	0	1
.	0	0	0	0	1	1	1	0

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Context Words |V|

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I	0	2	1	0	0	0	0	0
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deep	0	1	0	0	1	0	0	0
learning	0	0	0	1	0	0	0	1
NLP	0	1	0	0	0	0	0	1
flying	0	0	1	0	0	0	0	1
.	0	0	0	0	1	1	1	0

Source: https://bit.ly/3TGTaPH

Measure “co-occurrence” strength

Pointwise Mutual Information

Co-occurrence frequently | **Too frequent**

- **Word *w*, Context word *c***
  - $PMI(w, c) = \log \frac{p(w,c)}{p(w)p(c)}$

Measure "co-occurrence" strength

Pointwise Mutual Information

How many tokens does the corpus contain?

- Denote as N

How often does *w* occur?

- Denote as f(w)

- How often does *w* occur with *c* in its contexts?
  - Denote as f(w, c)
- How many tokens have *c* in their contexts?
  - Denote as f(c)

Word *w*, Context word *c*

$$PMI(w,c) = \log \frac{p(w,c)}{p(w)p(c)}$$

- $p(w,c) = \frac{f(w,c)}{N}$ $\quad p(w) = \frac{f(w)}{N}$ $\quad p(c) = \frac{f(c)}{N}$

PMI vs. Frequencies

"Co-occurrence" is defined by "object of" relations

	Frequency	PMI
"drink" vs. "bunch beer"	2	12.34
"drink" vs. "tea"	2	11.75
"drink" vs. "liquid"	2	10.53
"drink" vs. "champagne"	4	11.75
"drink" vs. "anything"	3	5.15
"drink" vs. "it"	3	1.25

(Lin, 1998)